

Friendly Order Types Through Incomparability Components

Finite ranks, successor defects,
and corrected disjoint-sum laws

Prepared for Vladimir Reshetnikov
19 September 2026

Abstract

The friendly order type of a well-partial order is the rank of its tree of bad sequences whose successive choices retain an incomparable witness. We attack the problem of computing this invariant structurally and compositionally. For every finite poset P , we prove

$$f(P) = |P| - c(P),$$

where $c(P)$ is the number of connected components of its incomparability graph. The proof gives an optimal witness sequence and a polynomial-time algorithm. For an infinite incomparability-connected well-partial order, we prove that the friendly order type is either its maximal order type or, when that type is a successor, its immediate predecessor. A canonical finite upper tail and a saturation test on the remaining core decide between the two cases. This yields exact disjoint-sum laws, including equality with maximal order type whenever both summands are infinite.

We also give a direct counterexample to a disjoint-sum identity printed in a 2023 paper. More strongly, two explicit countable orders have identical maximal order type, height, width, and friendly order type, but adjoining an incomparable singleton gives different friendly order types. Thus those four invariants do not support a general disjoint-sum calculation. The archive includes independent finite-rank and graph implementations, checked on all 101,660 naturally labeled posets with at most seven vertices.

Status. This is a research manuscript with detailed proofs, not a peer-reviewed or proof-assistant-verified publication. The finite and explicit counterexample arguments are elementary. The general transfinite classification uses an identified, previously published limit-floor bound. No claim of independently established priority is made. The broader problem of effective computation for arbitrary presented well-partial orders is not claimed to be settled.

Contents

1	The problem and the outcome	1
1.1	What is and is not an open-status claim	1
2	Definitions, ranks, and arithmetic	1
2.1	Friendly sequences	2
3	The incomparability graph and ordinal sums	3
3.1	A maximal-vertex lemma	4
4	An exact formula for every finite poset	4
4.1	Algorithm and certificate	5
5	The canonical finite upper tail	5
5.1	When full successor rank is possible	7
6	The successor-defect theorem	7
7	Exact disjoint-sum laws	9
8	Explicit counterexamples and independent calculations	9
8.1	An ordinal and finitely many incomparable points	10
8.2	A counterexample to the earlier printed identity	10
8.3	All four invariant values still do not suffice	11
9	Consequences and worked families	12
9.1	Finite multisets	12
9.2	Finite rectangular grids	12
9.3	Ordinal sums of finite blocks	12
9.4	Two successor behaviors	13
10	Computation and proof audit	13
10.1	What was checked by execution	13
10.2	What finite checking cannot establish	13
10.3	Dependency boundaries	14
11	What remains to investigate	14
A	Reproducing the artifacts	14

1 The problem and the outcome

Vialard’s 2024 thesis asks: “Can we compute the *for* compositionally?” [4, p. 109]. Here *for* means *friendly order type*. The invariant had appeared in the author’s MFCS 2023 work, where it determines the ordinal width of the multiset ordering [3, Theorem 3.4].

We extract a definite, tractable part of that research direction: find an intrinsic finite formula, determine the obstruction at infinite successor ranks, and understand disjoint sums. We also ask whether the standard three ordinal invariants, augmented by the friendly order type itself, contain enough information for such a calculation. This last question has a negative answer, witnessed by concrete countable orders below. The structural question has a positive answer in terms of incomparability components, maximal order types, and a single bit attached to an infinite core.

The most useful conclusions are the following. First, a finite friendly rank is simply the number of edges in a spanning forest of the incomparability graph. Second, an infinite connected component has a defect of at most one from maximal order type; the defect is completely classified. Third, disjoint sums of two infinite orders have no defect at all. Fourth, adding one isolated-in-the-order point can distinguish inputs that agree in all four familiar ordinal invariants.

These assertions are not all new definitions of special-purpose ranks: the rank being calculated is exactly Vialard’s friendly order type. We write $f(P)$ instead of $o_{\perp}(P)$ to keep the formulas readable.

1.1 What is and is not an open-status claim

The documented source poses a broad research question, not the exact finite equality proved here as a separately numbered conjecture. We do not relabel our extracted subproblem as a previously published conjecture. The current check covered the author’s thesis and publication page, the published MFCS paper, and targeted searches for later work on the friendly order type. No matching component-and-core formula was located. Search coverage and the absence of a located result do not establish priority.

There is a relevant version distinction. Proposition 4.1(2) of the 2023 published paper prints an unrestricted disjoint-sum identity. The 2024 thesis instead gives a sufficient limit-ordinal condition in Theorem 5.4.8(2). Section 8.2 supplies a direct counterexample to the earlier printed identity; it is not presented as contradicting the later conditional statement. The rest of the manuscript provides replacement formulas and makes clear which prior results are used.

2 Definitions, ranks, and arithmetic

All orders are sets, and the ambient foundational setting is ZFC. A *well-partial order*, or *wpo*, is a partial order having no infinite bad sequence. A finite sequence $x_0, \dots, x_{k-1}$ is *bad* when $x_i \not\leq x_j$ for every $i < j$. For $x \in P$, put

$$\uparrow_P x = \{y \in P : x \leq y\}, \quad P_x = P \setminus \uparrow_P x.$$

Thus selecting x in a bad sequence removes x and all its successors, not its predecessors. For a finite history s , the residual is $P_s = P \setminus \bigcup_{x \in s} \uparrow_P x$.

The rank of a node in a well-founded tree is the supremum of the ranks of its immediate successors plus one. A leaf has rank zero. The maximal order type $o(P)$ is the rank of the bad-sequence tree, equivalently the maximum order type of a linear extension. The existence

of a maximizing linear extension and the standard finite-sum laws are established wpo theory [1, 2]. In particular,

$$o(P) = \sup_{x \in P} (o(P_x) + 1), \quad o(A \sqcup B) = o(A) \oplus o(B). \quad (1)$$

Here $A \sqcup B$ is a disjoint sum: every element of A is incomparable to every element of B . By contrast, $A + B$ is an ordinal sum, with every element of A below every element of B .

The symbol $+$ denotes ordinary ordinal addition. The symbol $\oplus$ denotes Hessenberg, or natural, sum: write the two Cantor normal forms using a common decreasing list of exponents and add corresponding finite coefficients. In particular,

$$\alpha \oplus n = \alpha + n \quad (n < \omega), \quad \omega \oplus \omega = \omega \cdot 2.$$

Every infinite ordinal has a unique expression $\lambda + n$ with λ a nonzero limit ordinal and $n < \omega$. Whenever a finite correction is subtracted in this manuscript, it is subtracted *inside that finite coefficient*: $\lambda + (n - 1)$, not an unspecified ordinal subtraction.

2.1 Friendly sequences

Two elements x, y are *friends* when $x \perp y$, that is, they are incomparable. Following Vialard's definition, a bad sequence is *friendly*, or *open-ended*, if at each choice the selected element has a friend in the residual immediately before that choice [3, Definition 3.3]. A friend need not be selected later, and one friend may witness several successive choices.

The corresponding tree rank satisfies

$$f(P) = \sup_{\substack{x \in P \\ \exists y \in P (x \perp y)}} (f(P_x) + 1). \quad (2)$$

The empty supremum is zero. For a finite poset, this rank is exactly the maximum length of a friendly sequence. For an infinite wpo, taking the supremum of those finite lengths is generally *not* the tree rank.

Let $G(P)$ be the simple incomparability graph on P . Define

$$\text{str}(P) = \{x \in P : \exists y \in P (x \perp y)\}.$$

These are precisely the nonisolated vertices of $G(P)$.

Lemma 2.1. *For every wpo P ,*

$$f(P) = f(\text{str}(P)) \leq o(\text{str}(P)) \leq o(P).$$

The friendly rank is monotone under passage to an induced subposet.

Proof. A friendly sequence never selects a globally friendless element. Moreover, every friend of one of its selected elements lies in $\text{str}(P)$, since the friendship relation is symmetric. Consequently the friendly trees of P and $\text{str}(P)$ have exactly the same sequences. The first inequality follows by viewing this tree as a subtree of the bad-sequence tree on $\text{str}(P)$. For monotonicity, a friendly sequence in an induced subposet retains all its witness friends when considered in the larger order. The same subtree argument proves monotonicity of maximal order type. $\square$

We will use the following established consequence of the published bound for friendly rank. Its role is isolated so that the transfinite dependence is explicit.

Lemma 2.2 (Known limit-floor bound). *If $\text{o}(\text{str}(P)) = \lambda + n$, where λ is a nonzero limit ordinal and $n < \omega$, then*

$$\lambda \leq f(P) \leq \lambda + n.$$

In particular, $\text{o}(\text{str}(P)) = \lambda$ implies $f(P) = \lambda$.

This follows from the stronger bound $\lambda + \lfloor n/2 \rfloor \leq f(P)$ in [3, Theorem 4.5], also stated in [4, Theorem 5.4.4]. We use neither the unrestricted disjoint-sum identity in the former paper nor a deduction from that identity.

3 The incomparability graph and ordinal sums

Lemma 3.1 (Components are ordinally ordered). *The connected components of $G(P)$ are the factors in a canonical ordinal sum decomposition*

$$P = \sum_{\xi < \kappa} C_\xi$$

for an ordinal κ .

Proof. Elements in different components are comparable. If $a \perp b$ and z lies outside their component, the direction of comparison with z must agree: $a < z < b$ or $b < z < a$ would contradict $a \perp b$. Propagating this observation along finite incomparability paths shows that every comparison between two distinct components has the same direction. The resulting component order is linear. An infinite descending sequence of components would yield an infinite descending sequence in P by choosing representatives, contrary to the wpo property. Hence the component order is a well-order. $\square$

Theorem 3.2 (Transfinite ordinal-sum additivity). *For any ordinal-indexed family of wpos $(A_\xi)_{\xi < \kappa}$,*

$$f\left(\sum_{\xi < \kappa} A_\xi\right) = \sum_{\xi < \kappa} f(A_\xi). \quad (3)$$

Empty factors are permitted. The sum on the right is ordinary ordinal sum.

Proof. The ordinal sum is a wpo: a bad sequence cannot move to a strictly higher factor, and an infinite nonincreasing sequence of ordinal indices is eventually constant. An infinite bad sequence would therefore produce one inside a factor.

Prove the formula by induction on the maximal order type of the entire sum P . A selected $x \in A_\eta$ has a friend in P exactly when it has a friend in A_η . Its residual is

$$P_x = \left(\sum_{\xi < \eta} A_\xi\right) + (A_\eta)_x.$$

By (1), $\text{o}(P_x) < \text{o}(P)$, so the induction hypothesis applies to this residual sum. Write $\theta_\eta = \sum_{\xi < \eta} f(A_\xi)$. Then (2) gives

$$f(P) = \sup_{\substack{\eta < \kappa \\ x \text{ has a friend in } A_\eta}} (\theta_\eta + f((A_\eta)_x) + 1).$$

For a factor of positive friendly rank, continuity of addition in its right argument makes the supremum over x equal to $\theta_\eta + f(A_\eta)$. A zero-rank factor contributes no terms. Taking the

supremum over the positive-rank factors gives precisely the ordinal sum on the right of (3); intervening or terminal zero terms do not change it. If every factor has rank zero, both sides are zero. $\square$

The binary case is already stated in the literature [3, Proposition 4.1(1)]. The proof above also treats limit indices and zero-rank components, which will matter in applications.

3.1 A maximal-vertex lemma

A cut vertex of a connected graph is a vertex whose deletion leaves at least two connected components. An isolated singleton is not a cut vertex. The next elementary fact drives both the finite and infinite arguments.

Lemma 3.3 (At most one cut maximum). *In an incomparability-connected poset, at most one maximal element is a cut vertex of the incomparability graph.*

Proof. Suppose distinct maximal elements u, v were both cut vertices. Choose a component D of $G(P) - u$ not containing v . Connectivity of the original graph gives an $a \in D$ adjacent to u . There is no edge from a to v , so maximality of v gives $a < v$.

Similarly choose a component E of $G(P) - v$ not containing u , with a vertex $b \in E$ adjacent to v . Then $b < u$. Since $b \perp v$, the vertex b cannot belong to D : otherwise its edge to v would join D to v in $G(P) - u$. Thus a and b are in different components of $G(P) - u$, and must be comparable. If $a < b$, then $a < b < u$, contrary to $a \perp u$. If $b < a$, then $b < a < v$, contrary to $b \perp v$. Both alternatives are impossible. $\square$

Corollary 3.4. *A finite incomparability-connected poset with at least two elements has a maximal element whose deletion leaves the incomparability graph connected.*

Proof. A finite poset with a unique maximal element has a greatest element: every element lies below some maximal element. A greatest element is isolated in the incomparability graph. The present poset therefore has at least two maximal elements, and lemma 3.3 applies. $\square$

4 An exact formula for every finite poset

Write $c(P)$ for the number of components of $G(P)$ and set $c(\emptyset) = 0$.

Theorem 4.1 (Finite component formula). *For every finite poset P ,*

$$\boxed{f(P) = |P| - c(P)}. \quad (4)$$

There exists a friendly sequence attaining this length.

Proof. First let C be a connected nonempty component with r vertices. Any friendly sequence selects distinct elements. Its final selected element has an incomparable friend, and that friend survives the final deletion. Thus at least one element of C is left unselected, so its length is at most $r - 1$.

For the reverse inequality, if $r > 1$, use corollary 3.4 to select a maximal non-cut vertex x . It has a friend because the graph is connected and nontrivial. Maximality makes $C_x = C \setminus \{x\}$, still connected. Repeating this procedure leaves one vertex after exactly $r - 1$ friendly selections. The singleton case has rank zero, as required.

Finally decompose P into its finitely many ordinally ordered components. By theorem 3.2,

$$f(P) = \sum_C f(C) = \sum_C (|C| - 1) = |P| - c(P).$$

Witness sequences can be concatenated in descending component order. Once we move to a lower component, a deletion may also remove the unused singletons in higher components, but it does not remove the friends still needed in the current lower component. $\square$

Corollary 4.2 (Graphic interpretation). *For finite P , $f(P)$ equals the number of edges in any spanning forest of $G(P)$, equivalently the rank of its graphic matroid.*

Proof. A spanning tree of a component with r vertices has $r - 1$ edges. Summing over the components gives (4). $\square$

The finite invariant is consequently unchanged by reversing every order comparison: P and its dual have the same incomparability graph. Every value $r \in \{0, \dots, n - 1\}$ occurs on an n -element nonempty poset, for example on $\Gamma_{r+1} + [n - r - 1]$, where Γ_k is a k -element antichain and $[k]$ a k -element chain.

4.1 Algorithm and certificate

If the order relation is supplied as a reflexive transitive $n \times n$ matrix, form an incomparability edge exactly when neither direction of comparison holds. A graph traversal counts its components in $O(n^2)$ time. The value of the rank then follows from a single subtraction. A Hasse diagram input may first be transitively closed; the elementary dense method costs $O(n^3)$.

A witness is constructed component by component, from highest to lowest:

```
for C in incomparability_components_in_descending_order(P):
    while C contains more than one vertex:
        choose a maximal x in C for which G(C - {x}) is connected
        append x to the sequence
        C = C - {x}
```

At most two maximal candidates need to be tested at each stage, by lemma 3.3. Recomputing maxima and connectivity from a dense matrix therefore gives an $O(n^3)$ witness algorithm. A separate checker verifies the certificate by maintaining the actual residual of the original poset, checking a surviving friend, and deleting the selected element's upset. The archive implements this checker independently of the formula.

Example 4.3 (Three scalar invariants already fail in the finite case). The posets $\Gamma_2 + \Gamma_2$ and $[2] \sqcup [2]$ both have four elements, height two, and width two. Their incomparability graphs have respectively two components and one component. Their friendly ranks are therefore two and three. The stronger failure involving all four invariants in theorem 8.3 requires infinite examples.

5 The canonical finite upper tail

Definition 5.1. For a wpo P , define its *finite upper tail* and *core* by

$$T(P) = \{x \in P : |\uparrow_P x| < \omega\}, \quad P^\circ = P \setminus T(P).$$

The word “tail” does not assert that every core element lies below every tail element. The set $T(P)$ is an upset, but generally not an ordinal-sum factor of P .

Lemma 5.2 (Finiteness and stability). *The set $T(P)$ is a finite upset. For every finite subset $F \subseteq P$,*

$$T(P \setminus F) = T(P) \setminus F.$$

The core has no maximal elements.

Proof. If $x \leq y$ and $\uparrow x$ is finite, then $\uparrow y \subseteq \uparrow x$ is finite, proving upward closure. Suppose $T(P)$ were infinite. An infinite subset of a wpo contains an infinite strictly increasing sequence of distinct elements. To see this last fact, enumerate a countable subset and apply the infinite Ramsey theorem to the three possibilities for each pair of indices: increasing, decreasing, or incomparable. The last two homogeneous possibilities are excluded by the wpo property. An increasing sequence in $T(P)$ contradicts the finiteness of the upset of its first element. Thus $T(P)$ is finite.

Removing finitely many points from an infinite upset does not make it finite, and removing them from a finite upset leaves it finite. This proves the stability identity. Finally, each $x \in P^\circ$ has infinitely many successors in P ; all but finitely many remain in the core. In particular it has a strict successor there. $\square$

Lemma 5.3 (Removing a maximal element). *If x is a maximal element of a wpo P , then*

$$o(P) = o(P \setminus \{x\}) + 1.$$

Proof. Put $Q = P \setminus \{x\}$. A maximizing linear extension of Q , followed by x , is a linear extension of P , giving the lower bound. Forgetting all comparisons between x and Q gives the disjoint sum $Q \sqcup 1$. Every bad sequence in P is bad in that weaker order. Hence

$$o(P) \leq o(Q \sqcup 1) = o(Q) \oplus 1 = o(Q) + 1.$$

$\square$

Proposition 5.4 (The finite coefficient is intrinsic). *If P is infinite and $n = |T(P)|$, then*

$$o(P) = \lambda + n, \quad \lambda = o(P^\circ), \tag{5}$$

where λ is a nonzero limit ordinal. Thus n is exactly the finite coefficient at the end of the Cantor normal form of $o(P)$.

Proof. The core is infinite because only finitely many points have been removed. It has no maximal element. A linear extension of a poset with no maximal elements has no last element, so a maximizing linear extension of the core has nonzero limit type λ .

If the tail is nonempty, choose a maximal element in that finite upset. It is maximal in P because the tail is upward closed. Delete it and continue. After n such deletions the residual is exactly the core. Repeated use of lemma 5.3 gives (5). $\square$

This identifies the finite coefficient without selecting any particular maximal linear extension. For instance, the core of $\omega \sqcup 1$ is the chain ω , whereas the core of $(\omega \sqcup \omega) \sqcup 1$ consists of two incomparable infinite chains.

5.1 When full successor rank is possible

Lemma 5.5 (Full rank forces maximal moves). *Let $o(P) = \lambda + n$, with λ a nonzero limit ordinal and $0 < n < \omega$. If $f(P) = \lambda + n$, there is an eligible maximal element x such that*

$$f(P \setminus \{x\}) = o(P \setminus \{x\}) = \lambda + (n - 1).$$

Consequently full rank implies $f(P^\circ) = \lambda$.

Proof. A supremum equal to a successor ordinal is attained by one of its terms. Equation (2) therefore gives an eligible x with $f(P_x) = \lambda + (n - 1)$. Since $f(P_x) \leq o(P_x) < o(P)$, the maximal order type of P_x must have the same value.

Suppose $x < y$. No point of P_x lies above x or above y . A maximizing linear extension of P_x , followed by x and then y , is therefore a linear extension of the induced subposet $P_x \cup \{x, y\}$. It follows that

$$o(P) \geq o(P_x) + 2 = \lambda + (n + 1),$$

a contradiction. Thus x is maximal, so $P_x = P \setminus \{x\}$.

Repeat this argument n times. Every deleted point has a finite original upset: at its deletion all its original strict successors have already been deleted. The deleted set is therefore contained in $T(P)$ and has its full cardinality n . The remaining order is the core and has friendly rank λ . $\square$

6 The successor-defect theorem

The finite result loses one unit per connected component. In the infinite case that unit may disappear through ordinal limit behavior. The theorem below says precisely when it disappears.

Lemma 6.1 (The exceptional cut maximum). *Let C be an infinite incomparability-connected wpo with a unique maximal element u . If u is a cut vertex, then $C \setminus \{u\}$ has no maximal elements, and $o(C) = \lambda + 1$ for a nonzero limit λ .*

Proof. Every component of $G(C) - u$ has a vertex adjacent to u , since the original graph is connected. Suppose z were maximal in $C \setminus \{u\}$. If $z \perp u$, then z would be a second maximal element of C . Hence $z < u$. The component containing z must be the last component of $C \setminus \{u\}$ under lemma 3.1, since otherwise any point of a later component would be above z .

There are at least two components after deleting the cut vertex. Choose an earlier component D . Every $d \in D$ satisfies $d < z < u$, so no vertex of D is adjacent to u , a contradiction. Thus the remainder has no maximal element. Its maximal order type is a nonzero limit, as in proposition 5.4, and lemma 5.3 finishes the proof. $\square$

Lemma 6.2 (Deleting the tail without breaking connectivity). *Let C be infinite and incomparability-connected, with $o(C) = \lambda + n$ and $1 \leq n < \omega$. If $n \geq 2$, a maximal vertex can be deleted while preserving incomparability connectivity. Consequently one can delete $n - 1$ tail elements through friendly moves, leaving a connected order D with $o(D) = \lambda + 1$. The final tail point of D is also eligible, and deleting it leaves C° .*

Proof. There is a maximal element because $n > 0$ and $T(C)$ is nonempty. If there are two or more maximal elements, lemma 3.3 supplies a non-cut one. If there is only one, it cannot be a cut vertex when $n \geq 2$, by lemma 6.1 and uniqueness of the finite coefficient. Each chosen vertex has a friend because the current graph is infinite and connected. Deleting it

removes only that vertex and reduces the finite coefficient by one. Tail stability keeps the core unchanged.

After $n - 1$ deletions the remaining connected order has a tail consisting of one point. That point is maximal and still has a friend. Its deletion leaves the original core. No assertion that the core remains connected is needed or intended. $\square$

Definition 6.3 (Core defect bit). For an infinite wpo P , let $\lambda = o(P^\circ)$ and define

$$\varepsilon(P) = \begin{cases} 0, & o(\text{str}(P^\circ)) = \lambda, \\ 1, & o(\text{str}(P^\circ)) < \lambda. \end{cases}$$

The stripping operation is performed *after* the finite tail is removed.

Theorem 6.4 (Connected successor-defect classification). *Let C be a nonempty incomparability-connected wpo. If C is finite, then $f(C) = |C| - 1$. If C is infinite, put $\lambda = o(C^\circ)$ and $n = |T(C)|$. Then*

$$f(C) = \lambda + (n - \varepsilon(C)). \quad (6)$$

Here $\varepsilon(C) \leq n$. In particular, if $n = 0$ then $f(C) = o(C) = \lambda$. If $n > 0$, the only possible values are $\lambda + (n - 1)$ and $\lambda + n$.

Proof. The finite case is theorem 4.1. If $n = 0$, then $C^\circ = C = \text{str}(C)$, since an infinite connected graph has no isolated vertex. The limit-floor bound gives $f(C) = \lambda$ and $\varepsilon(C) = 0$.

Suppose $n \geq 1$. Apply lemma 6.2, making $n - 1$ friendly moves to a connected order D of maximal order type $\lambda + 1$. Since $\text{str}(D) = D$, lemma 2.2 gives $f(D) \geq \lambda$. Each previously made move adds one on the right to this lower bound. Hence

$$\lambda + (n - 1) \leq f(C) \leq \lambda + n. \quad (7)$$

There is no ordinal strictly between these two bounds.

If $\varepsilon(C) = 0$, then $o(\text{str}(C^\circ)) = \lambda$, and lemma 2.2 gives $f(C^\circ) = \lambda$. All n deletions of the tail described in lemma 6.2 are friendly, so $f(C) \geq \lambda + n$, establishing equality with the upper bound.

Conversely, equality with that upper bound forces $f(C^\circ) = \lambda$ by lemma 5.5. By lemma 2.1,

$$\lambda = f(C^\circ) \leq o(\text{str}(C^\circ)) \leq o(C^\circ) = \lambda.$$

Thus full rank is possible only when $\varepsilon(C) = 0$. If $\varepsilon(C) = 1$, the lower value in (7) must occur. $\square$

Remark 6.5 (Why connectedness cannot be dropped). An infinite chain has friendly rank zero and maximal order type equal to its own order type. Its defect can therefore be arbitrarily large. The connected theorem applies to each incomparability component, not to an arbitrary order without first decomposing it.

Corollary 6.6 (A structural formula for all wpos). *Let $P = \sum_{\xi < \kappa} C_\xi$ be its incomparability-component decomposition. For finite C_ξ , set $d_\xi = |C_\xi| - 1$. For infinite C_ξ , set*

$$d_\xi = o(C_\xi^\circ) + (|T(C_\xi)| - \varepsilon(C_\xi)).$$

Then $f(P) = \sum_{\xi < \kappa} d_\xi$.

Proof. Combine theorems 3.2 and 6.4. $\square$

The formula is intrinsic, but it is not a uniform decision procedure for arbitrary presentations of infinite orders. It still requires maximal order types of specified suborders and an equality test on the stripped core.

7 Exact disjoint-sum laws

The incomparability graph of $A \sqcup B$, for nonempty A, B , is connected: each vertex on one side is adjacent to every vertex on the other side. Its tail and core satisfy

$$T(A \sqcup B) = T(A) \sqcup T(B), \quad (A \sqcup B)^\circ = A^\circ \sqcup B^\circ. \quad (8)$$

These identities follow immediately from the equality of the principal upsets in a summand and in the disjoint sum.

Theorem 7.1 (Disjoint sums, all cases). *Let A, B be nonempty wpos.*

(i) *If both are finite, then $f(A \sqcup B) = |A| + |B| - 1$.*

(ii) *If both are infinite, then*

$$f(A \sqcup B) = o(A) \oplus o(B) = o(A \sqcup B). \quad (9)$$

There is no hypothesis that either maximal order type is a limit ordinal.

(iii) *If A is infinite and B is finite of size $m > 0$, write $o(A) = \lambda + n$, with $\lambda = o(A^\circ)$. Then*

$$f(A \sqcup B) = \lambda + (n + m - \varepsilon(A)). \quad (10)$$

The symmetric rule applies when A is finite and B infinite.

If one summand is empty, the result is simply the friendly rank of the other summand.

Proof. The finite case follows from connectedness and theorem 4.1.

If both summands are infinite, both cores in (8) are nonempty. Every core element has a friend in the other core, so the stripped core is the entire core. Its defect bit is zero. Apply theorem 6.4 and the standard maximal-order-type law (1).

In the mixed case, the core of the disjoint sum is precisely A° , its tail has $n + m$ points, and its core defect bit is $\varepsilon(A)$. Again apply theorem 6.4. $\square$

The mixed formula depends on the *cardinality* of the finite summand, not on its internal comparisons. This is stronger than merely giving bounds between consecutive ordinals. It also explains why information about the infinite core, rather than just the total rank, is necessary.

Corollary 7.2 (Finitely many nonempty summands). *For a finite disjoint family with at least two infinite summands, the friendly rank is the natural sum of the maximal order types of all the summands. With exactly one infinite summand A and a positive total of m elements in the finite summands, formula (10) applies. With all summands finite and at least two summands, the rank is one less than their total cardinality.*

Proof. With at least two infinite summands, their nonempty cores supply friends to each other, even after all finite tails and finite summands are removed. The core is therefore fully stripped. The other cases are applications of theorem 7.1 after grouping the finite summands. $\square$

8 Explicit counterexamples and independent calculations

The results in this section can be checked from the residual recurrence and finite arguments without using the limit-floor bound. This gives a separate check on the main examples.

8.1 An ordinal and finitely many incomparable points

Proposition 8.1. *For every ordinal α and every positive integer m ,*

$$f(\alpha \sqcup \Gamma_m) = \alpha + (m - 1). \quad (11)$$

In particular $f(\alpha \sqcup 1) = \alpha$.

Proof. For $\alpha = 0$ this is the finite antichain formula. For $m = 1$ and $\alpha > 0$, selecting $\beta < \alpha$ leaves $\beta \sqcup 1$; selecting the isolated point leaves the chain α , whose friendly rank is zero. Transfinite induction gives

$$f(\alpha \sqcup 1) = \sup(\{\beta + 1 : \beta < \alpha\} \cup \{1\}) = \alpha.$$

For $m \geq 2$ and $\alpha > 0$, induct on α , with an inner induction on m . A move in the antichain leaves $\alpha \sqcup \Gamma_{m-1}$ and contributes $\alpha + (m - 1)$ to the rank recurrence. A move at $\beta < \alpha$ in the chain contributes $\beta + m$. The latter values are at most $\alpha + (m - 1)$: for successor $\alpha = \gamma + 1$ their maximum is $\gamma + m$, and for limit α their supremum is α . The antichain move attains the required bound. $\square$

8.2 A counterexample to the earlier printed identity

Proposition 4.1(2) in the published 2023 paper states

$$f(A \sqcup B) = 1 + (o(A) - 1) \oplus (o(B) - 1). \quad (12)$$

Take $A = \omega \sqcup \omega$ and $B = 1$. We compute the actual left-hand side explicitly.

For $k \geq 1$, let $g_k = f(\omega \sqcup [k])$, and set $g_0 = 0$. A move in the infinite chain leaves a finite chain disjoint from $[k]$. The corresponding rank contributions have supremum ω , by the finite formula. A move at position j in the finite chain leaves $\omega \sqcup [j]$. Thus

$$g_k = \max\left(\omega, \max_{0 \leq j < k} (g_j + 1)\right) = \omega + (k - 1).$$

Selecting position k in either chain of $\omega \sqcup \omega$ leaves the other infinite chain disjoint from $[k]$. Hence

$$f(\omega \sqcup \omega) = \sup_{k < \omega} (g_k + 1) = \omega \cdot 2. \quad (13)$$

Now select the third, incomparable singleton. This is a friendly maximal move, and the residual is $\omega \sqcup \omega$. Therefore

$$f((\omega \sqcup \omega) \sqcup 1) \geq \omega \cdot 2 + 1.$$

The maximal order type of that order is $\omega \cdot 2 + 1$, so equality holds. On the other hand, the right-hand side of (12) is

$$1 + (\omega \cdot 2 - 1) \oplus (1 - 1) = 1 + \omega \cdot 2 = \omega \cdot 2,$$

using the infinite-ordinal subtraction convention also used in the source. The limit ordinal $\omega \cdot 2$ makes the discrepancy independent of any choice about how to decrement a successor ordinal's finite coefficient.

Proposition 8.2. *The unrestricted identity (12) fails for $A = \omega \sqcup \omega$ and $B = 1$. The actual value is $\omega \cdot 2 + 1$, not $\omega \cdot 2$.*

The 2024 thesis does not repeat (12); its stated disjoint-sum condition concerns two limit maximal order types. Accordingly this is an explicit correction to the printed formula, not a claim about whether the author had already recognized its limitation.

8.3 All four invariant values still do not suffice

Let

$$L = \sum_{k < \omega} \Gamma_2, \quad A = L + \Gamma_2, \quad B = (\omega + 1) \sqcup 1.$$

The order L consists of an ω -sequence of two-point antichain levels; every point at one level lies below every point at every later level.

The ordinary ordinal height $h(P)$ is the rank of the decreasing-sequence tree, and ordinal width $w(P)$ is the rank of the antichain-sequence tree. In the present examples every antichain has at most two elements and one has two, so the ordinal width is exactly two. A maximal chain in L has type ω , and a maximal chain in A or B has type $\omega + 1$. By theorems 3.2 and 4.1,

$$o(L) = 2 \cdot \omega = \omega, \quad f(L) = 1 \cdot \omega = \omega.$$

The following table records the resulting values; the two rightmost columns also locate the information that is lost by those four invariants.

Order	o	h	w	f	$ T $	ε
$A = L + \Gamma_2$	$\omega + 2$	$\omega + 1$	2	$\omega + 1$	2	0
$B = (\omega + 1) \sqcup 1$	$\omega + 2$	$\omega + 1$	2	$\omega + 1$	2	1

The formula for $f(B)$ is also an immediate instance of [proposition 8.1](#). Both orders have exactly two maximal elements, so adjoining the number of maximal elements to the list would not separate them either.

Theorem 8.3 (Failure of four-invariant compositionality). *The friendly order type of $P \sqcup Q$ is not determined by the values of o, h, w, f on the two summands. Specifically,*

$$f(A \sqcup 1) = \omega + 3, \quad f(B \sqcup 1) = \omega + 2, \tag{14}$$

although A and B agree in all four invariants.

Proof. For $A \sqcup 1$, delete the two points of its top Γ_2 one at a time, then delete the new incomparable singleton. All three moves are friendly: the singleton witnesses the first two, and a point of L witnesses the third. Only the chosen point is removed at each step. The residual is L , whose friendly rank is ω . Thus $f(A \sqcup 1) \geq \omega + 3$; its maximal order type is $\omega + 3$, proving equality.

On the other side, $B \sqcup 1 = (\omega + 1) \sqcup \Gamma_2$. Formula (11) gives $f(B \sqcup 1) = (\omega + 1) + 1 = \omega + 2$. The input invariant values coincide, including those of the common singleton summand, while the outputs differ. No rule depending only on those input values can therefore compute the output in every case. $\square$

The obstruction is not an exotic large ordinal. It occurs at the first infinite ordinal, with width bounded by two and only two maximal elements. In A , deleting the finite tail leaves an infinite supply of internal friendships. In B , deleting the finite tail leaves a linear order with none. The core defect bit records exactly this distinction.

9 Consequences and worked families

9.1 Finite multisets

For completeness, the multiset ordering $\mathcal{M}^r(P)$ consists of finite multisets over P , with

$$m \leq_r n \iff m = n \text{ or } \forall x \in m \setminus (m \cap n) \exists y \in n \setminus (m \cap n) (x < y).$$

Intersection and subtraction respect multiplicities. This is the Dershowitz–Manna-style multiset ordering, not the injective multiset embedding order. Vialard’s theorem gives

$$w(\mathcal{M}^r(P)) = \omega^{\mathfrak{f}(P)} \quad (15)$$

[3, Theorem 3.4]. Applying the formulas proved above yields

$$\boxed{w(\mathcal{M}^r(P)) = \omega^{|P|-c(P)} \text{ for finite } P.}$$

Even when P is empty, this is correct: its multiset order contains just the empty multiset and has width one. For two infinite wpos,

$$w(\mathcal{M}^r(A \sqcup B)) = \omega^{\mathfrak{o}(A) \oplus \mathfrak{o}(B)}.$$

The four-invariant example also yields $w(\mathcal{M}^r(A \sqcup 1)) = \omega^{\omega+3}$ but $w(\mathcal{M}^r(B \sqcup 1)) = \omega^{\omega+2}$.

9.2 Finite rectangular grids

Proposition 9.1. *For $m, n \geq 2$, with the product order on $[m] \times [n]$,*

$$\mathfrak{f}([m] \times [n]) = mn - 3, \quad w(\mathcal{M}^r([m] \times [n])) = \omega^{mn-3}.$$

Proof. The bottom corner $(0, 0)$ and top corner $(m-1, n-1)$ are isolated in the incomparability graph. Put $a = (0, n-1)$ and $b = (m-1, 0)$; they are adjacent. Every other noncorner vertex (i, j) is adjacent to a when $i > 0$ and $j < n-1$, or to b when $i < m-1$ and $j > 0$. The union of these two conditions excludes only the bottom and top corners. Thus all noncorner vertices form one component, giving exactly three components in total. Apply theorem 4.1 and (15). $\square$

If either factor is a singleton, the product is a chain and the rank is zero; the displayed grid formula must not be used in that case.

9.3 Ordinal sums of finite blocks

For finite nonempty blocks F_ξ ,

$$\mathfrak{f}\left(\sum_{\xi < \kappa} F_\xi\right) = \sum_{\xi < \kappa} (|F_\xi| - c(F_\xi)).$$

This realizes genuinely transfinite ranks from finite graph calculations. For instance,

$$\mathfrak{f}\left(\sum_{k < \omega} \Gamma_2\right) = \omega, \quad \mathfrak{f}\left(\sum_{\xi < \omega^2} \Gamma_3\right) = 2 \cdot \omega^2 = \omega^2.$$

A block that is a chain contributes zero, regardless of its finite length. The order of the blocks matters because ordinal addition is not commutative. For example,

$$\mathfrak{f}(L + \Gamma_3) = \omega + 2, \quad \mathfrak{f}(\Gamma_3 + L) = 2 + \omega = \omega.$$

These examples illustrate both finite losses and their absorption at a later limit.

9.4 Two successor behaviors

Both $\omega \sqcup 1$ and $(\omega \sqcup \omega) \sqcup 1$ are connected and have one point in the finite upper tail. In the first, the core is a chain, so the friendly rank is the predecessor ω of its maximal order type $\omega + 1$. In the second, the core remains connected and fully stripped, so the friendly rank retains the final successor:

$$f((\omega \sqcup \omega) \sqcup 1) = \omega \cdot 2 + 1.$$

This is precisely the distinction that a blanket “subtract one” rule cannot capture.

10 Computation and proof audit

10.1 What was checked by execution

The program code/verify.py enumerates naturally labeled posets on $\{0, \dots, n-1\}$: every strict comparison points from a smaller integer to a larger integer. The generator adds a new largest label, choosing any downset of the existing poset as its predecessor set. Every such poset is generated exactly once. Every finite isomorphism class has at least one natural labeling, but an isomorphism class can occur more than once.

For each generated poset, the program computes the rank independently by memoizing the defining residual recurrence on vertex masks. It then checks agreement with $n - c(P)$, constructs a witness by maximal non-cut deletions, checks the witness against the original residual semantics, and tests the at-most-one-cut-maximum assertion whenever the graph is connected.

Vertices	Naturally labeled posets checked	Connected graphs
0	1	0
1	1	1
2	2	1
3	7	4
4	40	27
5	357	275
6	4,824	4,070
7	96,428	86,278
Total	101,660	90,656

There were no failures. The verifier additionally checked all 25 grids with $2 \leq m, n \leq 6$ using the residual-rank oracle. The machine-readable output includes the full rank distributions, elapsed times, Python version, and execution timestamp. The associated unit tests also exercise scrambled vertex labels, invalid relations, empty orders, chains, antichains, ordinal sums, disjoint sums, and invalid witness sequences.

10.2 What finite checking cannot establish

The exhaustive computations do not verify an infinite tree rank. For example, finite suborders $[k] \sqcup [k]$ have friendly ranks $2k - 1$, whose supremum is only ω , while (13) gives $f(\omega \sqcup \omega) = \omega \cdot 2$. The additional successor in the three-summand counterexample is even less visible from finite truncation. The transfinite statements depend on the mathematical proofs, not on extrapolation from the finite data.

10.3 Dependency boundaries

Result	Dependencies and verification status
Finite component formula	Elementary component and cut-vertex lemmas; rank recurrence; exhaustive finite check.
Canonical finite tail	Standard wpo chain property, maximal linear extensions, natural-sum law, and the supplied deletion proof.
Connected classification	Supplied graph and tail arguments plus the previously published limit-floor bound.
General disjoint-sum laws	Connected classification and the explicit core identities.
Printed-identity counterexample	Direct recurrences for finite and infinite chains; no use of the disputed identity or the limit-floor bound.
Four-invariant counterexample	Finite formula, transfinite ordinal-sum additivity, and direct ordinal-plus-antichain calculation.
Multiset consequences	The supplied rank calculations and Vialard’s multiset-width theorem.

The new proofs are ordinary mathematical arguments rather than formal proof objects. In particular, a future formalization should preserve three points that are easy to obscure: every friend belongs to the *current* residual; every full-rank successor step is forced to select a maximal element; and the core need not remain connected after the last tail point is deleted.

11 What remains to investigate

The results answer the finite problem completely and give an exact structural classification for all incomparability components. They also supply a general disjoint-sum law once the core defect bit is available. The countable example proves that unrestricted four-invariant compositionality is not a viable substitute for that extra information.

A concrete next question is whether a useful class of presentations admits compositional calculation of

$$P \mapsto (\text{o}(P), |T(P)|, \text{o}(\text{str}(P^\circ))).$$

For example, one can ask this for orders built from ordinal constants, finite posets, ordinal sums, disjoint sums, and Cartesian products. The present paper settles the friendly-rank step once those data and the component decomposition are known, but does not provide every transition rule for that presentation language.

Another question is whether other common operations preserve a tractable core-saturation predicate. The disjoint sum of two infinite orders forces saturation automatically; an infinite order plus a finite disjoint summand preserves the defect bit of the infinite core. Products and higher-order constructions may require additional structural information. The finite grid calculation is an exact result for one product family, not a claimed classification of arbitrary products.

Finally, the elementary proofs of the two explicit counterexamples are small enough to make good formalization targets. A verified library of well-founded tree ranks and ordinal sums could certify them independently before attempting the more general connected-component classification.

A Reproducing the artifacts

The Python implementation uses only the standard library and is written for Python 3.9 or later. From the archive’s top directory, run:

```
python code/verify.py --max-n 7 --grid-max 6 \
    --output results/verification.json
python -m unittest discover -s code -v
```

The reference rank implementation can be exponential in the number of vertices; it is intentionally retained as an independent oracle. The component formula is the efficient method. Increasing the exhaustive bound increases the number of generated posets rapidly.

To rebuild the PDF, run `python build.py`, or use:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
    friendly_order_types.tex
```

The source uses the freely available `newpx` LaTeX packages for text and mathematics. Font binaries and third-party papers are not included. The archive also contains a source-status note and a short proof-audit checklist. The supplied `.bib` file and `.bbl` file allow both normal BibTeX builds and inspection of the rendered references.

References

- [1] Dick H. J. de Jongh and Rohit Parikh. Well-partial orderings and hierarchies. *Indagationes Mathematicae*, 39(3):195–207, 1977. doi:10.1016/1385-7258(77)90067-1.
- [2] Mirna Džamonja, Sylvain Schmitz, and Philippe Schnoebelen. On ordinal invariants in well quasi orders and finite antichain orders. In Peter Schuster, Monika Seisenberger, and Andreas Weiermann, editors, *Well Quasi-Orders in Computation, Logic, Language and Reasoning*, volume 53 of *Trends in Logic*, pages 29–54. Springer, 2020. doi:10.1007/978-3-030-30229-0_2.
- [3] Isa Vialard. Ordinal measures of the set of finite multisets. In Jérôme Leroux, Sylvain Lombardy, and David Peleg, editors, *48th International Symposium on Mathematical Foundations of Computer Science (MFCS 2023)*, volume 272 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 87:1–87:15. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2023. doi:10.4230/LIPIcs.MFCS.2023.87. Published version consulted 19 September 2026.
- [4] Isa Vialard. *Measuring Well Quasi-Orders and Complexity of Verification*. Phd thesis, Université Paris-Saclay, 2024. NNT 2024UPASG037. Author-hosted PDF, consulted 19 September 2026. Page references use printed page numbers.